METHANEX
CORPORATION

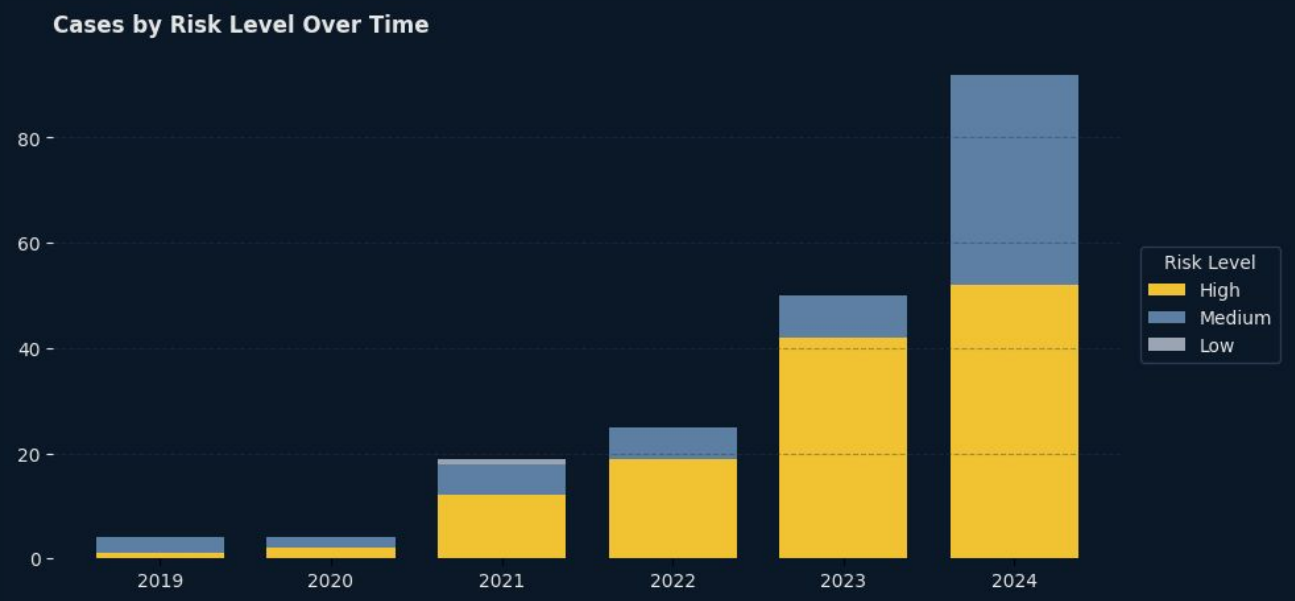
Safety Incident & Near-Miss Pattern Mining

Group 8: Safety in Your Area

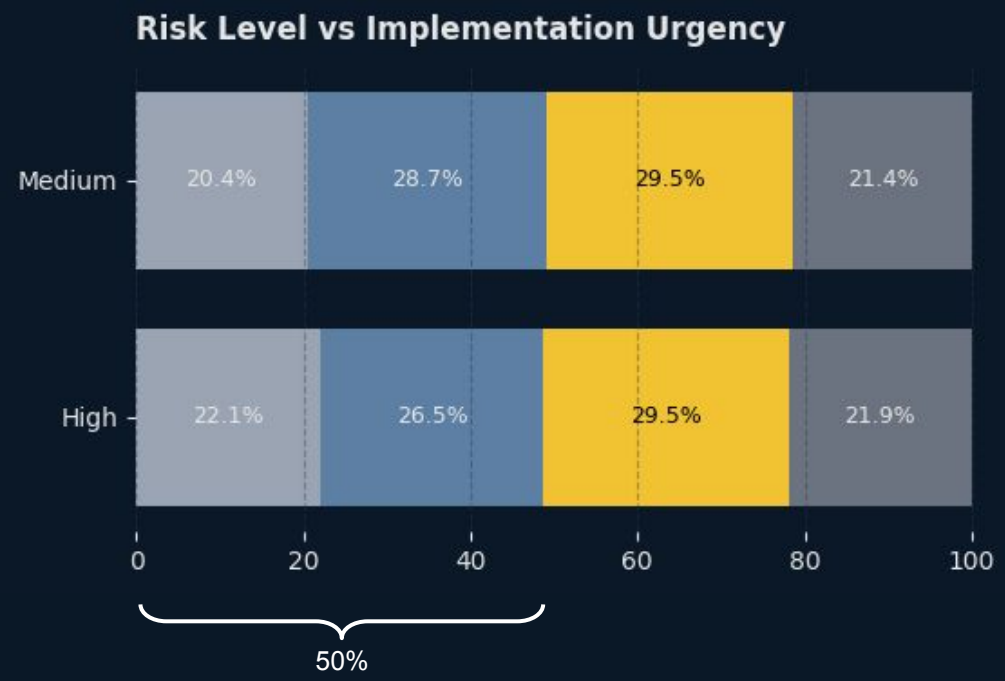
Google Cloud Hackathon, UBC, 2026

High Risk Concentration

Case volumes have risen over time, which may reflect improved reporting practices or increased incident occurrence. Regardless, High Risk remains the dominant category



Undifferentiated
Urgency

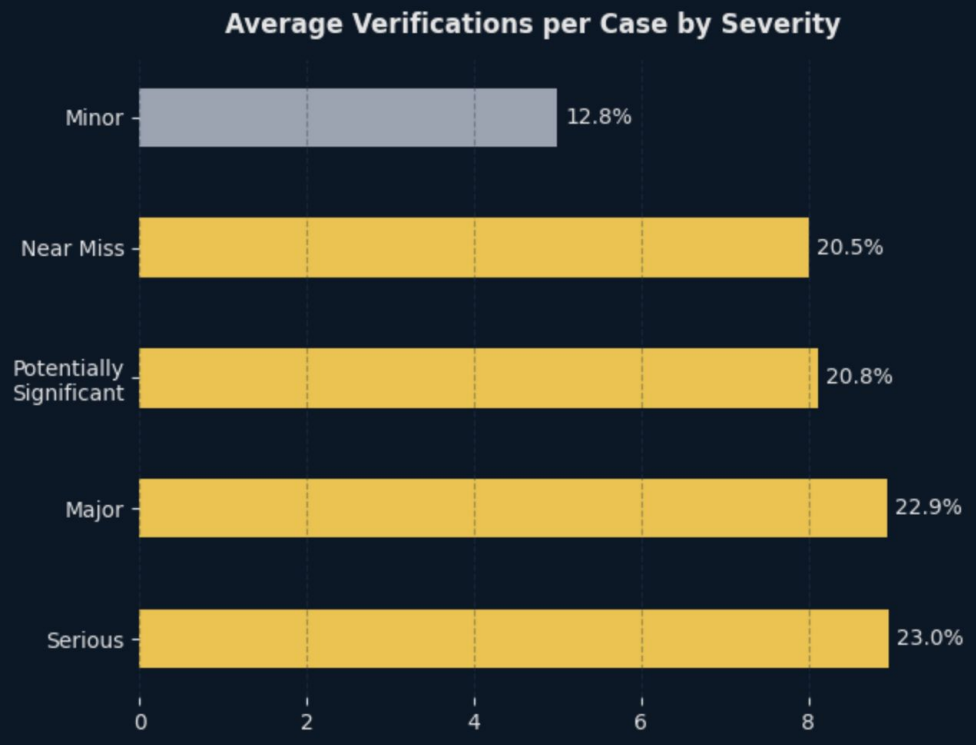


Timing Bucket
- Immediate
- 30–90 days
- <30 days
- >90 days

High & Medium risk cases have nearly same short-term action rates

Risk is not driving urgency

Undifferentiated follow-up depth



Verifications show how deep a case is investigated & formally validated after reporting

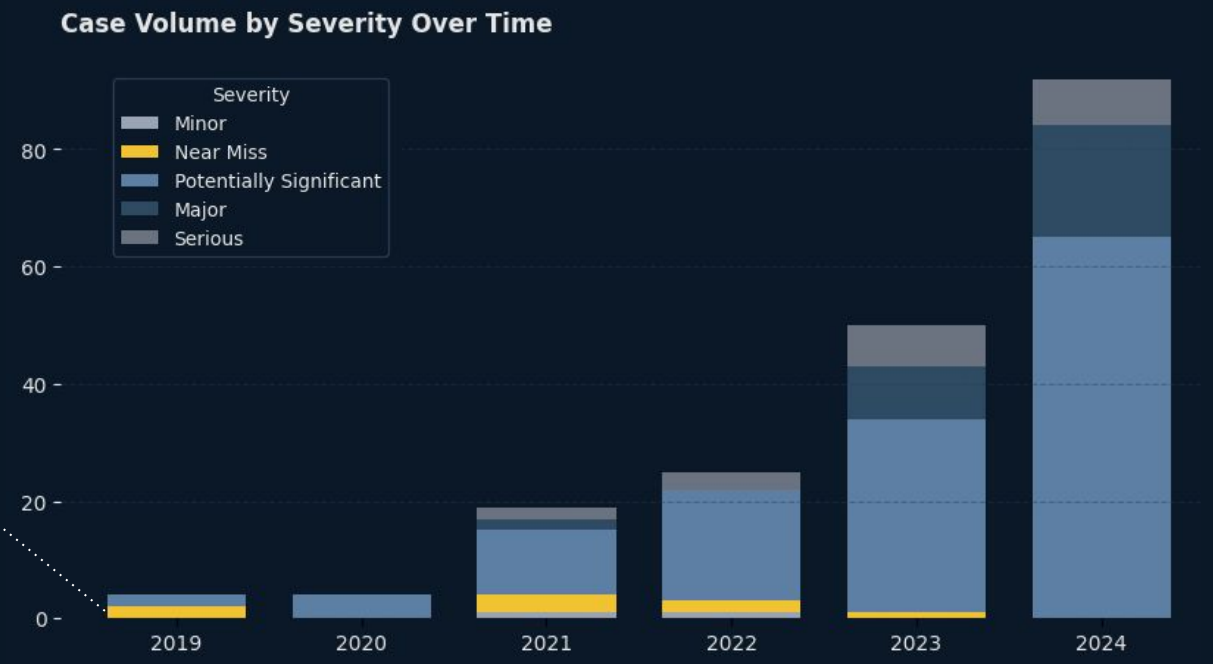
Flat verification strength across severity

→ Non-minor cases aren't getting stronger follow-up

Missed Early Signals

Near misses: critical early signals that help companies learn & prevent future cases

Rarity ≈ Limiting chances to proactively reduce risk



Non-standardized workflow, prolonged lead time

Lessons arrive only **after** incidents **occur**

Limit throughput as volume grows

*"Every incident is reviewed **case by case.**"*

*"It takes **1 week** to investigate and **2 weeks** to report."*

Delayed visibility **lowers** managers' **confidence**

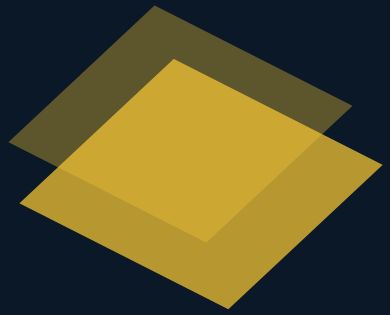
Difficult to scale without adding headcount

From Reactive System to Proactive Safety

Reactive → **Proactive**

- Missed Prevention
- Flat Risk Response
- Operational Bottleneck
- Delayed Visibility

Today



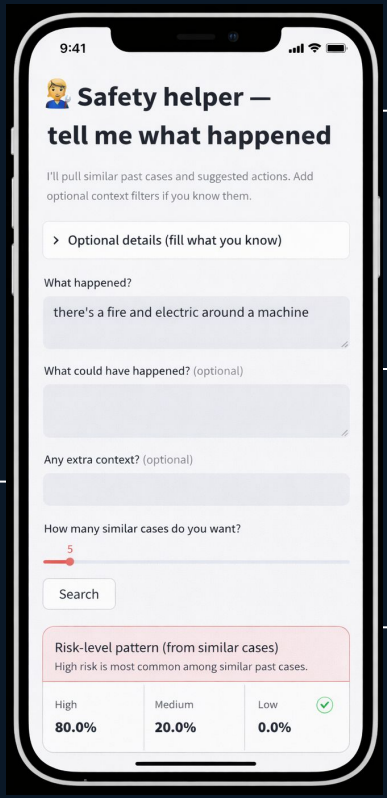
2-layer safety management system

- Early Warning & Pattern Detection
- Real-time Risk Recognition
- Faster Decision Support
- More visibility across organization

Tomorrow

Manager
Handbook App

From incident description to real-time guidance in seconds. Learn more



Smart incident input

Managers enter what they see for quick guideline

Pattern recognition

Automatically identify similar scenarios & past incidents

Risk estimation

Instantly estimates likelihood of severity outcomes based on historical patterns.

Action playbook

Suggest proven actions from similar past cases

My Guideflow

👷 Safety helper — tell me what happened

I'll pull similar past cases and suggested actions. Add optional context filters if you know them.

> Optional details (fill what you know)

What happened?

Enter details in the What happened? field to describe the incident.

What could have happened? (optional)

Any extra context? (optional)

How many similar cases do you want?

5

Guideflow.

Cluster 0

Information & Confidential Exposure Risk
Focus: employee data, document handling, workspace hazards
Typical drivers: verification gaps, sensitive materials, shared spaces
Risk impact: privacy exposure, electrical hazards, slip & workplace injuries

Cluster 1

AI & Automation Control Reliability Risk
Focus: AI recommendations, sensor logic, automated decisions
Typical drivers: incorrect model outputs, control misalignment
Risk impact: equipment damage, operational disruption, safety escalation

Cluster 2

Isolation & Permit Compliance Risk
Focus: maintenance work, line isolation, contractor activities
Typical drivers: incomplete isolation, valve handling, timing gaps
Risk impact: vapor release, exposure risk, process disruption

Cluster 3

Stored Energy & Pressure Release Hazards
Focus: hydraulic, pneumatic, compressed air & pressure systems
Typical drivers: bleed failures, actuator release, pressure buildup
Risk impact: projectile hazards, injection injuries, blast force exposure

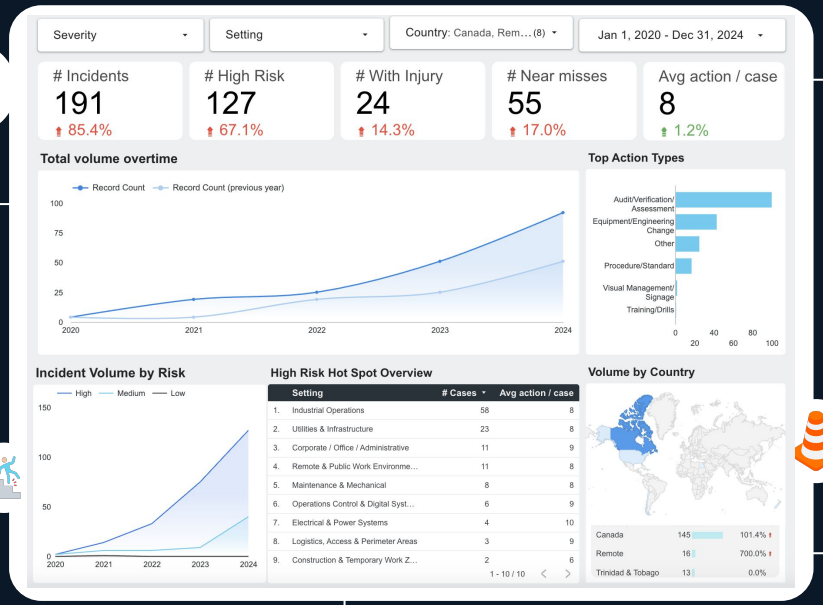
Cluster 4

Access Control & Privilege Misuse Risk
Focus: security access, network control, contractor permissions
Typical drivers: badge access gaps, system override, cyber access
Risk impact: process safety incidents, environmental & compliance risk

Executive Safety Dashboard

Real-time safety intelligence for leadership teams

Learn more



Trend Tracking

Total incidents over time, split by risk level

KPI Summary Stats

incidents, high-risks, injuries, near misses, avg. actions per case

Volume by Country

Interactive map to visualize geographic incident distribution

High Risk Hotspots

Settings ranked by # case, with ave # actions per case

Business Impact & Strategic Value

01

Faster Time-to-Action

Real-time risk classification & guidance **e**nables **same-day prevention**

📉 **Time reduction:** *weeks → seconds*

02

Low-Cost, Scalable Deployment

~$1 per user per day
~$300/month for a 10-person department

💰 Enterprise-grade insight at minimal marginal cost

03

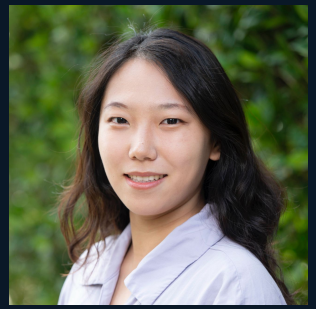
Leadership-Ready Visibility

Faster leadership decisions via **cross-site comparison** & **accountability**

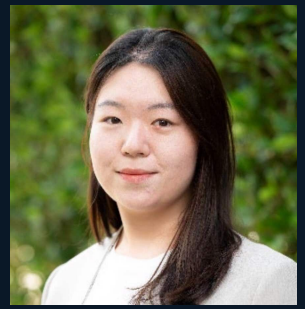
📊 **C**learer ownership, smarter prioritization

Thank you.

Thao Do



Jasmine Fu



Liz Ji

Jessica Shi

Appendix 💡

METHANEX

Estimated Deployment Cost

Scaling Safety Intelligence Across Teams

Group 8

Cost Projection Snapshot

Observed usage (pilot)

- 4 users
- 3 days
- $12.21 total

Scaled Estimate (Department Level)

Per person per day	~$1.02
10-person department	~$10/day
Monthly estimate	~$290~$315

Enterprise deployment
remains low-cost while scaling impact.

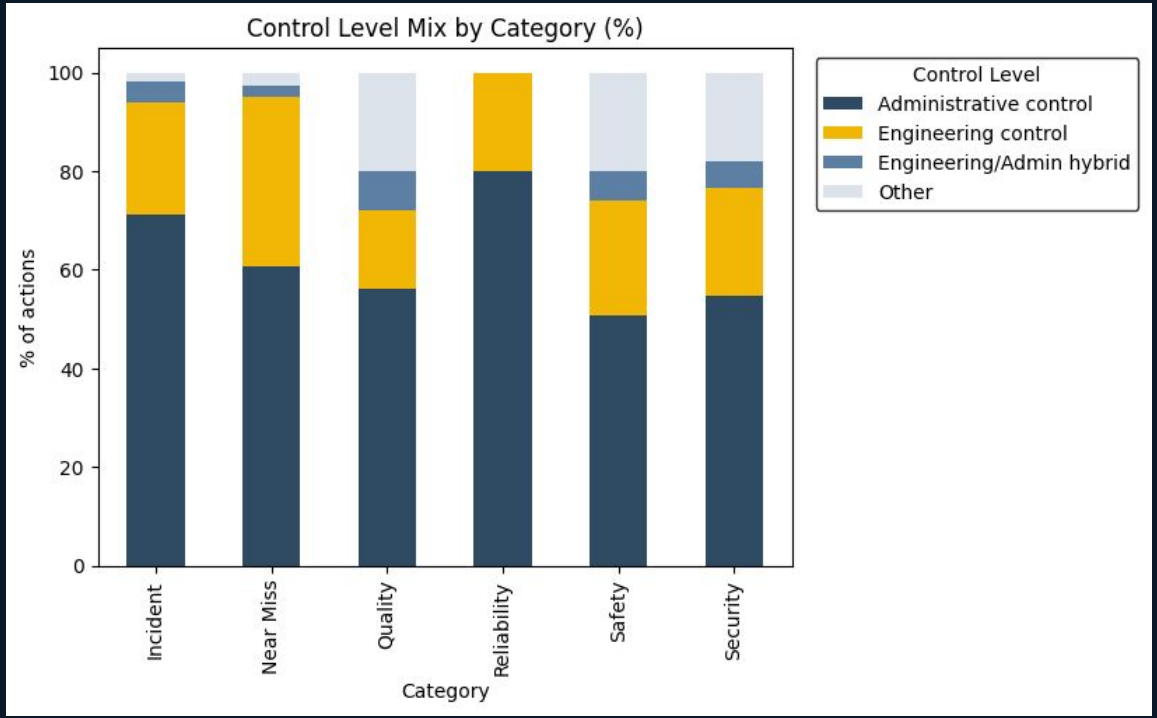
Why Cost Stays Efficient

- Cloud usage scales with activity, not headcount
- Pattern recognition reduces manual investigation time
- Early detection prevents costly incidents
- No additional infrastructure required

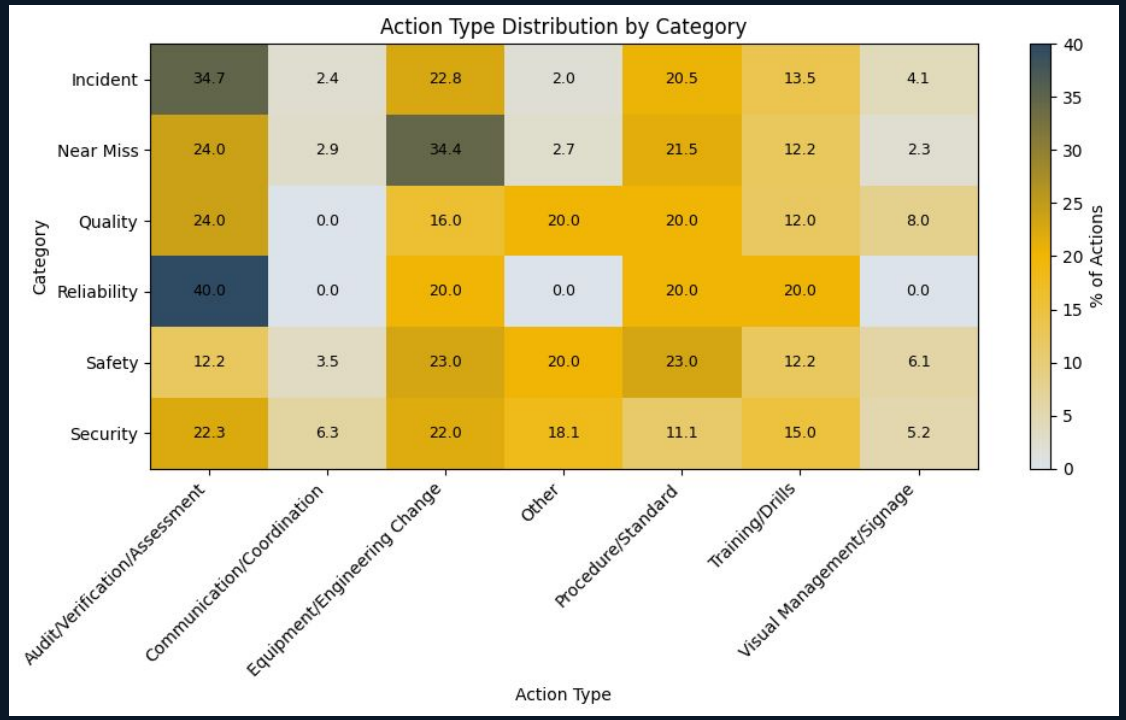
Executive Takeaway

High safety impact.
Minimal operational cost.
Scales instantly.

Over half of actions are administrative, signaling a dependence on procedural fixes over structural prevention



Near misses rely more on engineering fixes, while incidents lean on audit-based controls, signaling reactive action patterns



Underutilized prevention window

>60%

Major & Serious cases resulted in no injury.

Risks surfaced after incidents happen, instead of being captured earlier through near-miss learning